

PROJECT WORK

GROCERY STORE SYSTEM

Python Programming Project

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Submitted as part of Python Programming / Problem Solving coursework.

1. Project Title

Grocery Store System

2. Introduction

The Grocery Store System is a simple console-based Python application for basic grocery shopping and billing. It allows the user to view grocery items, add or remove items from a cart, calculate the total bill, apply a discount, and display the final amount.

3. Objectives

- Create a simple menu-driven grocery shopping program.
- Practice Python dictionaries, functions, loops and conditional statements.
- Calculate item-wise and total prices automatically.
- Apply discount rules to the bill.
- Handle invalid input safely.

4. Technologies Used

Python, Visual Studio Code, dictionaries, functions, loops, conditional statements, input/output and exception handling.

5. Main Features

- Display available grocery items and prices.
- Add items and quantities to the cart.
- Remove items or quantities from the cart.
- Calculate the total amount.
- Apply discount according to the purchase amount.
- Display the final bill.

6. Working of the Project

The program first stores grocery products and their prices. An empty cart is created. A menu is repeatedly displayed to the user. The user can add an item, remove an item, display the bill, or exit. Functions divide these operations into manageable parts. The final amount is calculated from price × quantity and the applicable discount is deducted.

7. Algorithm

1. Start.
2. Create the grocery item-price dictionary and an empty cart.
3. Display the main menu.
4. Read the user's choice.
5. Add or remove items according to the choice.
6. Calculate the total and discount when the bill is requested.
7. Display the final bill.
8. Repeat until Exit is selected.
9. Stop.

8. Python Concepts Used

- Dictionary – stores grocery products and prices.
- Functions – separate program operations.
- if/elif/else – menu and validation logic.
- while loop – keeps the menu running.
- for loop – processes cart/store items.
- try/except – handles invalid numeric input.

9. Advantages

The program is simple, menu-driven, easy to understand, and demonstrates practical use of basic Python programming concepts. It automatically calculates bills and reduces manual calculation.

10. Limitations

The product list is fixed in the program. It does not use a database or permanent storage and has a console-based interface.

11. Future Scope

The project can be extended with database storage, inventory management, customer details, invoice generation, a graphical interface, and online payment functionality.

12. Conclusion

The Grocery Store System is a practical beginner-level Python project that combines dictionaries, functions, loops, conditions, calculations and input validation. It demonstrates how fundamental programming concepts can be used to build a small real-world application.

13. Project Source Code

```
# Simple Grocery Store System
# Features: add item, remove item, calculate bill, apply discount, display bill

# Items available in the store (name: price per unit)
store_items = {
    "milk": 60,
    "bread": 40,
    "rice": 90,
    "eggs": 70,
    "sugar": 45,
    "apple": 120,
}

# Cart stores item name and quantity
cart = {}

def show_store_items():
    print("\n--- Store Items ---")
    for name, price in store_items.items():
        print(name.capitalize(), "- Rs", price)

def add_item():
    show_store_items()
    name = input("\nEnter item name to add: ").lower()

    if name not in store_items:
        print("Sorry, item not available.")
        return

    try:
        qty = int(input("Enter quantity: "))
    except ValueError:
        print("Please enter a valid number.")
        return

    if qty <= 0:
        print("Quantity must be more than 0.")
        return

    if name in cart:
        cart[name] = cart[name] + qty
    else:
        cart[name] = qty
    print(qty, name, "added to cart.")

def remove_item():
    if len(cart) == 0:
        print("Cart is empty.")
        return

    name = input("Enter item name to remove: ").lower()

    if name not in cart:
        print("Item is not in the cart.")
        return

    try:
        qty = int(input("Enter quantity to remove: "))
    except ValueError:
        print("Please enter a valid number.")
        return

    if qty >= cart[name]:
        del cart[name]
        print(name, "removed from cart.")
    elif qty > 0:
        cart[name] = cart[name] - qty
        print(qty, name, "removed from cart.")
    else:
        print("Invalid quantity.")

def calculate_total():
    total = 0
    for name, qty in cart.items():
        total = total + store_items[name] * qty
```

```

    return total

def get_discount(total):
    # Simple discount rules
    if total >= 1000:
        return 10 # 10 percent
    elif total >= 500:
        return 5 # 5 percent
    else:
        return 0

def display_bill():
    if len(cart) == 0:
        print("Cart is empty. Nothing to bill.")
        return

    print("\n===== BILL =====")
    print("Item      Qty   Price   Amount")
    print("-----")
    for name, qty in cart.items():
        price = store_items[name]
        print(name.capitalize().ljust(10), str(qty).ljust(5),
              str(price).ljust(7), price * qty)

    total = calculate_total()
    discount_percent = get_discount(total)
    discount_amount = total * discount_percent / 100
    final_amount = total - discount_amount

    print("-----")
    print("Total      : Rs", total)
    print("Discount (" + str(discount_percent) + "%) : Rs", discount_amount)
    print("Final Amount : Rs", final_amount)
    print("=====")

def main():
    while True:
        print("\n===== GROCERY STORE =====")
        print("1. Add item to cart")
        print("2. Remove item from cart")
        print("3. Display bill")
        print("4. Exit")

        choice = input("Enter your choice (1-4): ")

        if choice == "1":
            add_item()
        elif choice == "2":
            remove_item()
        elif choice == "3":
            display_bill()
        elif choice == "4":
            print("Thank you for shopping!")
            break
        else:
            print("Invalid choice. Try again.")

main()

```